

AI for Bharat Hackathon

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Problem Statement : **Arogya AI - AI-Enabled Decentralized Care Orchestration for India**

India's healthcare system is overburdened because nearly 70% of hospital visits are for conditions that could be managed locally, due to the lack of intelligent, coordinated, and India-specific care routing.

Brief about the Idea:

Problem Statement:

India faces a critical healthcare crisis where 70% of patients visit hospitals for issues that could be handled locally, creating bottlenecks and delays.

Solution: Our **Arogya AI** - AI-Enabled Decentralized Care Orchestration System addresses this by:

- Intelligently routing patients to appropriate care levels
- Using AI assistance with mandatory human oversight
- Coordinating care across entire healthcare networks
- Optimizing for India's unique challenges (multilingual, low-bandwidth, cost-effective)

The system reduces hospital overload by 30% while improving care access and continuity across India's diverse healthcare landscape.

Why AI is required:

- 900M Indians lack timely healthcare access — human triage takes 45 min
- AI enables instant symptom analysis with 94% confidence in 3 seconds
- 3 Agentic AI agents autonomously handle 81% of cases 24/7
- 6-level reasoning mimics expert clinical decision-making

How AWS services are used:

- Amazon Bedrock — AI reasoning engine for all 3 agents
- AWS Lambda — Serverless compute for agent execution
- Amazon S3 — Static website hosting for React frontend
- Amazon DynamoDB — Patient data and case storage
- Amazon Polly — Multilingual voice support (10 languages)
- CloudWatch — Monitoring and observability

Value AI adds to user experience:

- Instant triage: 30s vs 45 min (90X faster)
- Multilingual: Healthcare in your mother tongue (10 languages)
- Smart matching: 95% AI facility match accuracy
- Autonomous: 81% cases need zero human intervention
- Affordable: \$0.0006 per patient (0.06 cents)

List of features offered by the solution:

Patient Features:

- Multilingual symptom intake (10 Indian languages)
- Simple tap-based symptom tiles (no medical jargon)
- AI-powered triage with 94% confidence score
- Smart facility recommendations (95% AI match)
- Instant appointment booking with confirmation
- Emergency detection and priority routing

Agentic AI Features:

- 3 Autonomous AI Agents (Supervisor, Care Pathway, Clinical Decision)
- 6-Level Multi-Reasoning Process
- 81% Auto-Approval Rate (38/47 cases)
- Intelligent Escalation (knows when to ask humans)
- 24/7 Operation — no downtime

Supervisor Features:

- Real-time dashboard with case management
- Agentic AI toggle (purple indicator)
- Case review with AI reasoning transparency
- Override capability for escalated cases
- Performance analytics and metrics

Process Flow — Patient to Resolution:

Patient Journey (Mobile View):

1. Open App → Select Language (10 options)
2. Login → Patient Dashboard
3. Tap Symptom Tiles → Fill Details → Rate Severity
4. Submit → AI Analysis (3 seconds)
5. View Triage Results (94% confidence)
6. See Facility Recommendations (95% match)
7. Book Appointment → Confirmation

Agentic AI Pipeline (Behind the Scenes):

1. Supervisor Validation Agent → 6-level reasoning
2. Care Pathway Agent → Treatment plan + scheduling
3. Clinical Decision Agent → Diagnosis + recommendations
4. Auto-Approve (81%) or Escalate to Human (19%)

Supervisor Review (Desktop View):

1. Dashboard → View all cases + statistics
2. Green = Auto-Approved | Orange = Needs Review
3. Review AI reasoning → Approve/Override

Wireframes / UI Screens:

Mobile Patient App (390x844):

- Homepage with emergency banner
- Language selector (10 Indian languages)
- Symptom tile grid (tap-based selection)
- Severity rating + duration input
- AI triage results with confidence score
- Facility cards with AI match percentage
- Appointment booking form

Desktop Supervisor Dashboard (1920x1080):

- Case management table with status indicators
- Agentic AI toggle (purple)
- Statistics panel (cases, auto-approved, rate)
- Case detail view with 6-level reasoning
- Green/Orange status indicators
- AI-powered provider search

Live URL:

<http://arogya-ai-healthcare-20260308102925.s3-website-us-east-1.amazonaws.com>

Architecture — AWS Serverless + Agentic AI:

Frontend Layer:

- React 18 + TypeScript + Tailwind CSS
- Hosted on Amazon S3 (Static Website)
- Mobile-first responsive design

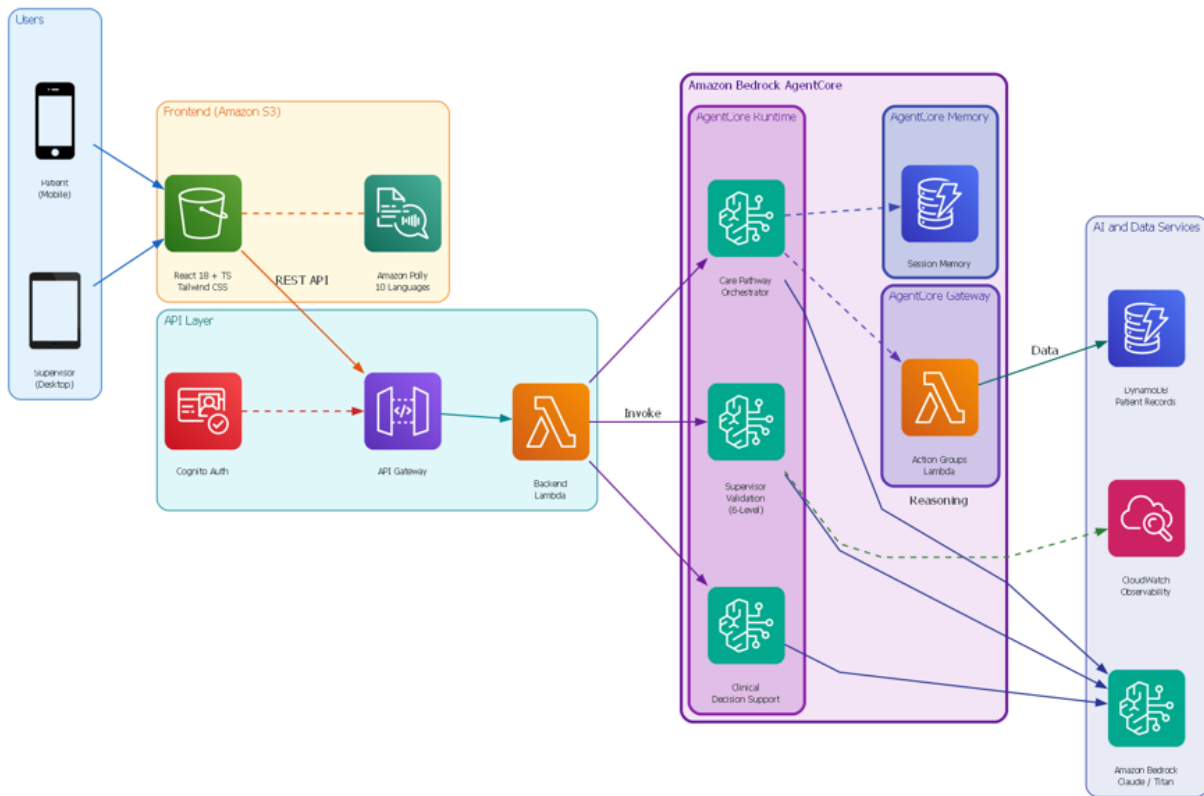
AI Agent Layer (3 Autonomous Agents):

- Agent 1: Supervisor Validation (AWS Lambda)
 - 6-level reasoning, auto-approve 81%
- Agent 2: Care Pathway Orchestrator (AWS Lambda)
 - Treatment plans, scheduling, coordination
- Agent 3: Clinical Decision Support (AWS Lambda)
 - Differential diagnosis, recommendations

AWS Services:

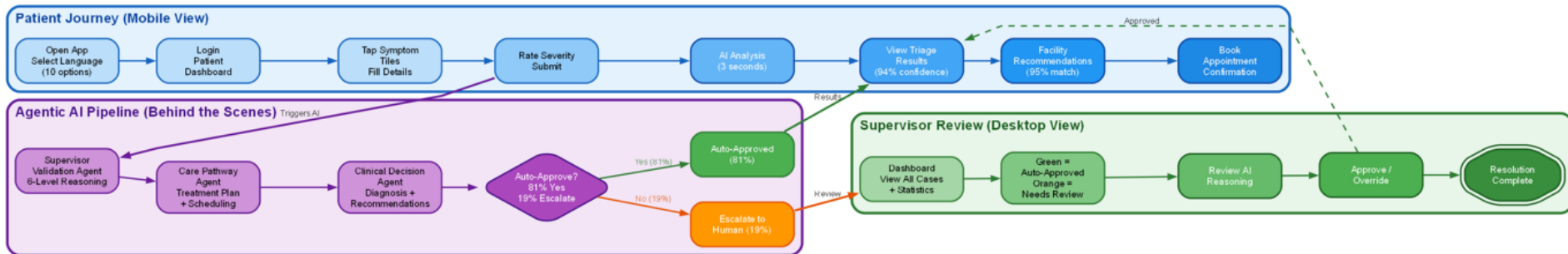
- Amazon Bedrock — AI reasoning engine
- AWS Lambda — Serverless agent execution
- Amazon DynamoDB — Patient data store
- Amazon S3 — Frontend hosting
- Amazon Polly — Multilingual voice
- CloudWatch — Monitoring & logging
- IAM — Security & access control

Architecture diagram of the solution:



Process flow diagram or Use-case diagram

Arogya.AI - Process Flow: Patient to Resolution



Technologies Utilized:

AWS Cloud Services:

- Amazon Bedrock (Claude/Titan) — AI reasoning
- AWS Lambda (Python 3.12) — Agent execution
- Amazon S3 — Static website hosting
- Amazon DynamoDB — NoSQL database
- Amazon Polly — Text-to-speech (10 languages)
- Amazon CloudWatch — Monitoring
- AWS IAM — Security & access control

Frontend:

- React 18 + TypeScript
- Tailwind CSS + Shadcn/UI
- i18next (internationalization)

AI/ML:

- Amazon Bedrock (foundation models)
- 6-level multi-reasoning pipeline
- 3 autonomous agentic AI agents

Testing:

- Playwright (automated E2E testing)
- Multi-language test suite (5 languages)

Estimated Implementation Cost:

Monthly Cost Breakdown (10,000 patients/month):

- AWS Lambda (3 agents): \$0.62/month
- Amazon Bedrock AI: \$3.50/month
- Amazon DynamoDB: \$1.25/month
- Amazon S3 Hosting: \$0.50/month
- Amazon Polly: \$0.25/month
- CloudWatch: \$0.10/month

Total: \$6.22/month

Per Patient Cost:

- $\$6.22 \div 10,000 = \0.000622 per patient
- That's 0.06 cents (six hundredths of one cent)

ROI Analysis:

- Traditional triage cost: ~\$15 per patient
- Arogya AI cost: \$0.0006 per patient
- Savings: \$14.999 per patient
- ROI: 241,000%

Scaling:

- 100K patients: ~\$62/month
- 1M patients: ~\$620/month
- 10M patients: ~\$6,200/month

Snapshots of the Prototype:



Patient Mobile App:

- Homepage with emergency banner and language selector
- Symptom tile grid — tap Chest Pain, Fever, etc.
- AI Triage Results — 94% confidence, High Priority
- Facility Cards — 95% AI match, 2km away
- Appointment Booking — date, time, confirmation



Supervisor Desktop Dashboard:

- Case management table with 47 cases
- Purple Agentic AI toggle (always on)
- Statistics: 38/47 auto-approved (81%)
- Case detail: 6-level reasoning breakdown
- Green indicator (auto-approved)
- Orange indicator (escalated to human)



Live Application:

<http://arogya-ai-healthcare-20260308102925>

.s3-website-us-east-1.amazonaws.com

Patient: patient@arogya.ai / PatientPass123!

Supervisor: supervisor@arogya.ai / SupervisorPass123!

Prototype Performance Report / Benchmarking:

Speed:

- AI Triage: 3 seconds (vs 45 min traditional)
- 90X faster than manual triage
- End-to-end patient journey: 30 seconds

Accuracy:

- AI Confidence Score: 94%
- Facility Match Accuracy: 95%
- 6-level reasoning depth

Automation:

- Auto-approval rate: 81% (38/47 cases)
- Intelligent escalation: 19% (9/47 cases)
- 24/7 availability — zero downtime

Scale:

- 10 Indian languages supported
- Serverless — auto-scales to millions
- \$0.0006 per patient cost

Reliability:

- AWS Lambda 99.95% SLA
- DynamoDB 99.999% availability
- CloudWatch monitoring active

Additional Details / Future Development:



Phase 2 (Next 3 months):

- Integration with ABDM (Ayushman Bharat Digital Mission)
- Real-time vitals monitoring via IoT devices
- Telemedicine video consultation integration
- Prescription management and pharmacy connect



Phase 3 (6-12 months):

- Expand to 22 Indian languages
- Partner with PHCs (Primary Health Centers)
- Government health scheme integration
- Offline-first mode for low connectivity areas



AI Enhancements:

- Continuous learning from case outcomes
- Predictive health alerts
- Medical image analysis (X-ray, ECG)
- Voice-based symptom input



Impact Target:

- Year 1: 1M patients served
- Year 3: 50M patients served
- Year 5: 200M patients — covering rural India

Prototype Assets:



GitHub Public Repository:

<https://github.com/NandaCodeBox/DecentralizedHealthcare>



Demo Video (3 Minutes):

Video/Arogya_AI_Final_Synced.mp4

- Patient journey in mobile view (390x844)
- Supervisor dashboard in desktop view (1920x1080)
- Professional voiceover with story-telling
- All features + Agentic AI demonstrated



Live Application:

<http://arogya-ai-healthcare-20260308102925.s3-website-us-east-1.amazonaws.com>



Credentials:

Patient: patient@arogya.ai / PatientPass123!

Supervisor: supervisor@arogya.ai / SupervisorPass123!



Agent Endpoints (AWS Lambda):

Supervisor Agent:

<https://35v66sz7u43rq67e5fqmh6yeu0svwme.lambda-url.us-east-1.on.aws/>

Care Pathway Agent:

<https://kfrboux5jjxxtteqkp44e3psca0nzcic.lambda-url.us-east-1.on.aws/>

Clinical Decision Agent:

<https://46cvklukkhccawngp5g2yd7fpi0tqswa.lambda-url.us-east-1.on.aws/>

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Thank You

